Lec 4 : 9.1 - 10.3

概念自检

9.1 - 9.6

- binary relation
  - the graph of function
  - relation on a set
- connection matrix
- directed graph / digraph
  - vertex
  - edge / arc
- properties of binary relations
  - reflexive
  - irreflexive
  - symmetric
  - antisymmetric
  - transitive
- combining relations
  - composition
- power of relation
- inverse relation
- the properties of relation operations
- closure
  - reflexive / symmetric / transitive closure
- warshall's algorithm
- equivalence relation
  - reflexive / symmetric / transitive
- partition
- partial ordering
  - reflexive / antisymmetric / transitive
  - poset / partial ordering set
- comparable / incomparable
- total order / linear order / chain
- lexicographic order
- Hasse diagram
- chain / antichain
- maximal / minimal element

- greatest / least element
- upper / lower bounds
- least upper bounds / greatest lower bounds
- well-ordered set
- lattice
- topological sorting

10.1 - 10.3

- graph
  - vertices
  - edges
  - endpoints
- simple graph
- multigraph
- pseudograph
- directed graph / digraph
  - simple directed graph
  - directed multigraph
- degree of vertex
- **the handshaking theorem**
- degree 为奇数的点有偶数个；出度之和等于入度之和
- complete graph / K_n
- cycles / C_n
- wheels / W_n
- n-Cubes / Q_n
- bipartite graphs
  - 充分必要条件?
- regular graph
- subgraph
  - proper subgraph
  - spanning subgraph
- union of graphs
- adjacency matrices
- incidence matrices
- isomorphism of graphs

习题整理

9.1 - 9.6

1. 2018秋冬 期末

   ( ) Let P be the set of all propositional formulas with variables p, q, r, and let R be the equivalence relation on P defined by $\varphi R \psi$ iff $\varphi \leftrightarrow \psi$, for $\varphi, \psi \in P$. Then the number of equivalence classes of P on R is 256.

2. 2019秋冬 期末

   5. (12 pts) Let F be the set of all propositional formulas involving variables $p_1, p_2, \cdots, p_n$. Define a binary relation $\mathbf{R}$ on F:

   For all $A, B \in F$, $A\mathbf{R}B$ if and only if $A \leftrightarrow B$ is a tautology.

   (a) Prove that the binary relation R on F is an equivalence relation.

   (b) Give a description of the equivalence classes.

   (c) How many different equivalence classes of R are there?

3. 2018秋冬 期末

   ( ) There is a binary relation R on set A such that R is both an equivalence relation and a partial order on A.

4. 2018秋冬 期末

   ( ) Suppose $(A, \preceq)$ is a finite nonempty poset. Then A has a minimal element.

5. 2018秋冬 期末

   Define a binary relation R on $\mathbb{R} \times \mathbb{R}$ by $(x, y)R(a, b)$ if and only if $y - b = 5(x - a)$.

   (a) Show that R is an equivalence relation on $\mathbb{R} \times \mathbb{R}$.

   (b) Find the equivalence classes of $(0, 0)$ and $(1, -2)$.

   (c) Describe the geometric meaning of the equivalence class $[(a, b)]$.

6. 2019秋冬 期末

   ( ) Let R be a binary relation. If R is symmetric and transitive, then R is reflexive.

7. 2019秋冬 期末

( ) Let $(S, \preceq)$ be a partially ordered set. If there is a unique minimal element e of S, then e is the least element of S.

8. 2019秋冬 期末

   8. (10 pts) Let S be a set having n elements. Let H be the Hasse diagram for the partial ordering $\{(A, B) \mid A \subseteq B\}$ on the power set of S. Let f_n denote the number of edges in a Hasse diagram representing a set with n elements.

   (a) Compute f_0, f_1, f_2 and f_3.

   (b) Find a recurrence relation for f_n, and justify your answer.

9. 24春夏 llf 第三次小测

   ( ) (1) The set of intervals $[k, k+1]$, $k = \cdots, -2, -1, 0, 1, 2, \cdots$ is the partition of the set of all integers.

10. 24春夏 llf 第三次小测

( ) (2) There are two different equivalence relations on a set with two elements.

11. 24春夏 llf 第三次小测

( ) (3) Let R be a relation on the set A. R equals its transitive closure if and only if R is transitive.

12. 24春夏 llf 第三次小测

( ) (4) The poset $(\{1, 2, 3, 4, 6, 8\}, |)$ is a lattice, where $x \mid y$ denotes x divides y.

13. 24春夏 llf 第三次小测

(1) Suppose $A = \{1, 2\}$, there are __ relations on the set A, there are __ partial orderings.

14. 24春夏 llf 第三次小测

(5) Find the smallest partial order on $\{1, 2, 3\}$ that contains $(1, 1)$, $(3, 2)$, $(1, 3)$.

15. 25春夏 llf 第三次小测

( ) (1) The symmetric closure of a transitive relation is still transitive.

16. 25春夏 llf 第三次小测

(1) Suppose $|A| = n$, there are __ distinct total order relations on A.

17. 25春夏 llf 第三次小测

Find the transitive closure of R if $M_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$.

18. 25春夏 图灵 第三次小测

Find the transitive closure of R if the adjacency matrix M_R is $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$.

19. 25春夏 llf 第三次小测

Let A be the set of bit strings of length not exceeding three, where λ is a bit string of length zero. Suppose that R is a relation on the set A, where $(a, b) \in R$ if and only if a is a prefix of b. For example, 00 is a prefix of 001, and 01 is not a prefix of 001.

(1) Show that R is a partial order.

(2) Draw the Hasse diagram for the partial order.

(3) Find its minimal elements and the greatest element.

20. 25春夏 图灵 第三次小测

(6) Find the smallest equivalence relation on $\{1, 2, 3, 4, 5\}$ that contains $(1, 2)$, $(1, 4)$ and $(3, 5)$.

21. 25春夏 图灵 第四次小测

( ) If a relation R is not transitive then R^2 is also not transitive.

22. 25春夏 图灵 第四次小测

Let $A = \{(a, b) \mid a, b \in \mathbb{Z}^+\}$. Define the relation R on A as
$R = \{((a, b), (c, d)) \mid ad + a = bc + c\}$.

(1) Determine whether R is an equivalence relation.

(2) Prove your answer.

10.1 - 10.3

23. 25 llf 第三次小测

Give a recurrence relation for e_n ， where e_n is the number of edges of the graph Q_n.

24. 25 llf 第三次小测

( ) K_3 has 6 non-isomorphic subgraphs.

25. 25 turing quiz3

There are ___ non-isomorphic simple directed graphs with 3 vertices and 2 edges.

26.25 zyw 第三次小测

Suppose that $S_1, S_2, \ldots, S_n$ is a collection of subsets of a set S where n is a positive integer. A system of distinct representatives (SDR) for this family is an ordered n-tuple $(a_1, a_2, \ldots, a_n)$ with the property that $a_i \in S_i$ for $i = 1, 2, \ldots, n$ and $a_i \neq a_j$ for all $i \neq j$.

Use Hall's Marriage theorem to show that a collection of finite subsets $S_1, S_2, \ldots, S_n$ of a set S has a SDR $(a_1, a_2, \ldots, a_n)$ if and only if $\forall I \subseteq \{1, 2, \ldots, n\}$, $\left|\bigcup_{i \in I} S_i\right| \geq |I|$.

注：Hall婚配定理的内容是：对于一个二分图 $G = (V, E)$，它的二分为 (V_1, V_2)，当且仅当对于 V_1 的所有子集 A，满足 $|N(A)| \geq |A|$ 时，该二分图有从 V_1 到 V_2 的完全匹配。

27. 证明：在至少含有两个顶点的简单图中，一定有两个顶点的度相同

28. 对哪些n的值来说，下列图是二分图?

- K_n
- C_n
- W_n
- Q_n

29. 一个图的度序列是由该图的各个顶点的度按非递增序列排列的序列。

判断下列序列是否是成图的（即能否构成一个简单图的度序列）

- 5,4,3,2,1,0
- 6,5,4,3,2,1
- 2,2,2,2,2,2
- 3,3,3,2,2,2
- 3,3,2,2,2,2
- 1,1,1,1,1,1
- 5,3,3,3,3,3
- 5,5,4,3,2,1
- 3,3,3,3,2
- 5,4,3,2,1
- 4,4,3,2,1
- 4,4,3,3,3
- 3,2,2,1,0
- 1,1,1,1,1

> 注：Havel-Hakimi 定理可以判断一个给定的有限非负整数序列是否是可图化的（即能否构成一个简单无向图的度序列），感兴趣可自行了解。